TITLE PAGE

TEAM MEMBERS

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PROJECT METADATA

- **Problem Statement ID: SIH26-PS153**
- **Problem Statement Title: Graph World Model for Predictive Cyber Attack Forecasting**
- **Theme: Cybersecurity & Critical Information Infrastructure**
- **PS Category: Software**
- **Team ID & Name: SIH2026 PS153 APEX / APEX CODERS**

IDEA TITLE: VANGUARD-GWM PLATFORM

- **Proposed Solution: Vanguard-GWM**

  Models computer networks as evolving temporal graphs to forecast multi-stage APT attacks before host compromise occurs.

- **Core Pipeline & Workflow**

  • Ingests dual-level telemetry (flow aggregates + transport header geometry). • Constructs dynamic network graph G_t using Polars and NetworkX. • Employs PyTorch Geometric GATv2 spatial encoders and Causal Temporal Transformers to predict state transitions and execute K-step autoregressive rollouts.

- **How it Addresses the Core Challenge**

  • Replaces static, reactive classifiers (Random Forest / LSTM) that trigger post-facto. • Delivers +180s early warning during the reconnaissance phase, stopping multi-stage APTs prior to lateral movement.

- **Innovation & Uniqueness**

  • True predictive world model rather than a reactive signature filter. • Payload-agnostic telemetry fusion with built-in Explainability (XAI) and What-If containment simulation.

TECHNICAL APPROACH

1. Telemetry Ingestion
Scapy, PyShark,
Polars (Δt=1-5s)

2. Graph Construction
Dynamic Graph G_t
(Nodes & Edges)

3. World Model Rollout
GATv2 + Causal
Transformer (K-steps)

4. SDN Actuation
OpenFlow 1.3
QUARANTINE / DROP

• Methodology & Implementation Flow:

• Continuous streaming telemetry via Scapy/PyShark/Polars. Extracts rolling window metrics to construct dynamic graph G_t. • PyTorch Geometric GATv2 encoder compresses network topology into latent state z_t; Causal Temporal Transformer learns transition dynamics P(z_{t+1}|z_t) [World Model core]. • Performs K-step forward simulation predicting risk score and staging automated OpenFlow 1.3 SDN DROP rules via a Streamlit SOC interface.

• Technology Stack

• Frontend/UI: Streamlit, Plotly | Backend: Python FastAPI | AI/ML Framework: PyTorch, PyTorch Geometric | Database: NetworkX, Polars Dataframes | Networking API: OpenFlow 1.3

FEASIBILITY AND VIABILITY

POTENTIAL CHALLENGES & RISKS

- **Telemetry Parsing Latency:**
  Slow PCAP processing during live demo. Fix: Pre-extract feature matrices ahead of time and use Polars for high-speed streaming dataframes.
- **Graph Over-smoothing:**
  GAT layers losing separation on large networks. Fix: Use a 2-3 layer GAT with InfoNCE contrastive loss over subgraphs.
- **Autoregressive Latent Drift:**
  Drift accumulating beyond K>5 steps. Fix: Cap rollout horizon at K=3-5 time windows and add temporal consistency regularizers.

FEASIBILITY & PROTOTYPE SCOPE

- **Verified Performance:**
  Fully prototyped with sub-50ms inference latency (36.4 ms), operating entirely offline/air-gapped without cloud API dependencies.
- **Prototype Scope Pipeline:**
  Telemetry Ingestion ➔ OCSF/Feature Slicing ➔ GATv2 Encoder ➔ World Model Core ➔ SDN Enforcer.
- **Deployment Readiness:**
  Pinned Docker environment, automated unit testing suite, and complete GitHub repository structure.

APEX CODERS

IMPACT AND BENEFITS

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POTENTIAL IMPACT ON TARGET AUDIENCE

- **Enterprise SOCs & CII:**
  Shifts cybersecurity operations completely from reactive alert triage to proactive, predictive defense.
- **Eliminates Alert Fatigue:**
  Groups thousands of fragmented log entries into a single, cohesive kill-chain trajectory narrative.
- **National & Defense Impact:**
  Secures critical national infrastructure against stealthy, zero-day Advanced Persistent Threats (APTs).

BENEFITS & QUANTIFIED OUTCOMES

- **Proactive Lead Time:**
  +180s to +600s early warning during initial reconnaissance with 97.9% multi-stage APT recall[cite: 1] (2.0% FPR[cite: 1]).
- **Decision Support:**
  What-If simulation lets analysts test host quarantine actions (ΔRisk = -61.1%[cite: 1]) before acting.
- **Economic Value:**
  Protects critical infrastructure, minimizes breach costs, and prevents analyst burnout.

APEX CODERS

RESEARCH AND REFERENCES

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PRIMARY BENCHMARK DATASETS

- CSE-CIC-IDS2018 and CTU-13 network traffic datasets.
- Attack timelines divided into temporal windows to build sequential network states and ground-truth state transitions.

EVALUATION METHODOLOGY

- Held-out attack variants to measure unseen-attack generalization.
- Benchmarked against Logistic Regression & Random Forest baselines across Precision, Recall, F1, and Predictive Lead Time[cite: 1].

CORE ACADEMIC REFERENCES

- Ha, D., & Schmidhuber, J. (2018). 'World Models.' arXiv:1803.10122.
- Velickovic, P., et al. (2018). 'Graph Attention Networks.' ICLR.

PROJECT REPOSITORY & DOCS

- GitHub Repository: github.com/raghuraj2008-cu/Vanguard-GWM[cite: 1]
- IEEE Conference Manuscript: Vanguard_GWM_IEEE_Conference_Paper.pdf[cite: 1]